

GEOMETRY

In most competitive exams, the problems relating to Geometry usually cover triangles, quadrilaterals and circles. Even though polygons with more than four sides are also covered, the emphasis on such polygons is not as much as it is on triangles and circles. In this chapter, we will look at some properties as well as theorems and riders on parallel lines, angles, triangles (including congruence and similarity of triangles), quadrilaterals, circles and polygons.

ANGLES AND LINES

An angle of 90° is a right angle; an angle less than 90° is an acute angle; an angle between 90° and 180° is an obtuse angle; and an angle between 180° and 360° is a reflex angle.

The sum of all angles made on one side of a straight line AB at a point O by any number of lines joining the line AB at O is 180° . In Fig. 4.01 below, the sum of the angles u, v, x, y and z is equal to 180° .

When any number of straight lines meet at a point, the sum of all the angles around that point is 360° . In Fig. 4.02 below, the sum of the angles u, v, w, x, y and z is equal to 360° .

Fig. 4.01

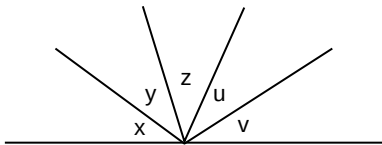
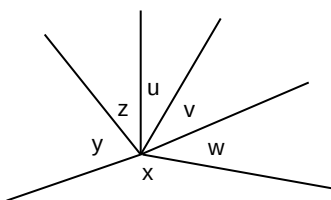


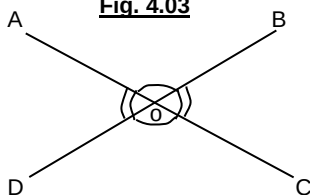
Fig. 4.02



Two angles whose sum is 90° are said to be complementary angles and two angles whose sum is 180° are said to be supplementary angles.

When two straight lines intersect, vertically opposite angles are equal. In Fig. 4.03 given below, $\angle AOB$ and $\angle COD$ are vertically opposite angles and $\angle BOC$ and $\angle AOD$ are vertically opposite angles. So, we have $\angle AOB = \angle COD$ and $\angle BOC = \angle AOD$.

Fig. 4.03



Two lines which make an angle of 90° with each other are said to be PERPENDICULAR to each other.

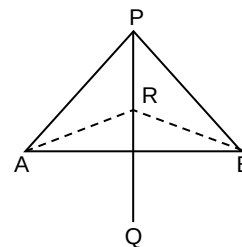
If a line/line segment l_1 passes through the mid-point of another line segment l_2 , the line/line segment l_1 is said to be the BISECTOR of the line segment l_2 , i.e., the line segment l_2 is divided into two equal parts.

If a line l_1 is drawn at the vertex of an angle dividing the angle into two equal parts, the line l_1 is said to be the ANGLE BISECTOR of the angle. Any point on the angle bisector of an angle is EQUIDISTANT from the two arms of the angle.

If a line l_1 is perpendicular to line segment l_2 as well as passes through the mid-point of the line segment l_2 , the line l_1 is said to be the PERPENDICULAR BISECTOR of the line segment l_2 .

Any point on the perpendicular bisector of a line segment is EQUIDISTANT from both ends of the line segment.

Fig. 4.04

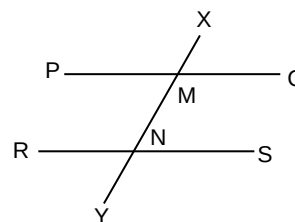


In Fig 4.04, line PQ is the perpendicular bisector of line segment AB. A point P on the perpendicular bisector of AB will be equidistant from A and B, i.e., $PA = PB$. Similarly, for any point R on the perpendicular bisector PQ, $RA = RB$.

PARALLEL LINES

When a straight line cuts two or more lines in the same plane, the line is called the TRANSVERSAL. When a transversal XY cuts two parallel lines PQ and RS [as shown in Fig. 4.05], the following are the relationships between various angles that are formed. [M and N are the points of intersection of XY with PQ and RS respectively].

Fig. 4.05

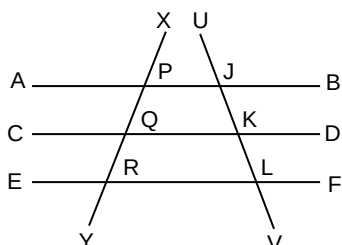


- Alternate angles are equal, i.e.
 $\angle PMN = \angle MNS$ and $\angle QMN = \angle MNR$
- Corresponding angles are equal, i.e.
 $\angle XMQ = \angle MNS$; $\angle QMN = \angle SNY$;
 $\angle XMP = \angle MNR$; $\angle PMN = \angle RNY$
- Sum of interior angles on the same side of the transversal is equal to 180° , i.e.
 $\angle QMN + \angle MNS = 180^\circ$ and $\angle PMN + \angle MNR = 180^\circ$
- Sum of exterior angles on the same side of the transversal is equal to 180° , i.e.
 $\angle XMQ + \angle SNY = 180^\circ$, and $\angle XMP + \angle RNY = 180^\circ$

RATIO OF INTERCEPTS

If three or more parallel lines make intercepts on a transversal in a certain proportion, then they make intercepts in the same proportion on any other transversal as well. In Fig. 4.06, the lines AB, CD and EF are parallel and the transversal XY cuts them at the points P, Q and R. If we now take a second transversal, UV, cutting the three parallel lines at the points J, K and L, then we have $PQ/QR = JK/KL$.

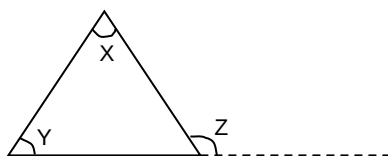
Fig.4.06



Note: If three or more parallel lines make equal intercepts on one transversal, they make equal intercepts on any other transversal as well.

TRIANGLES

Fig. 4.07



Sum of the three angles of a triangle is 180°

The exterior angle of a triangle (at each vertex) is equal to the sum of the two opposite interior angles. (Exterior angle is the angle formed at any vertex, by one side and the extended portion of the second side at that vertex).

A line perpendicular to a side and passing through the midpoint of the side is said to be the **perpendicular bisector** of the side. It is not necessary that the perpendicular bisector of a side should pass through the opposite vertex in a triangle in general.

The perpendicular drawn to a side from the opposite vertex is called the **altitude** to that side.

The line joining the midpoint of a side with the opposite vertex is called the **median** drawn to that side. A median divides the triangle into two equal halves as far as the area is concerned.

An equilateral triangle is one in which all the sides are equal (and hence, all angles are equal, i.e., each of the angles is equal to 60°). An isosceles triangle is one in which two sides are equal (and hence, the angles opposite to them are equal). A scalene triangle is one in which no two sides are equal.

In an isosceles triangle, the unequal side is called the **BASE**. The angle where the two equal sides meet is called the **VERTICAL ANGLE**. In an isosceles triangle, the perpendicular drawn to the base from the vertex opposite

the base (i.e., the altitude drawn to the base) bisects the base as well as the vertical angle. That is, the altitude drawn to the base will also be the perpendicular bisector of the base as well as the angle bisector of the vertical angle. It will also be the median drawn to the base.

In an equilateral triangle, the perpendicular bisector, the median and the altitude drawn to a particular side coincide and that will also be the angle bisector of the opposite vertex. If a is the side of an equilateral triangle, then its altitude is equal to $\frac{\sqrt{3}}{2}a$.

Sum of any two sides of a triangle is greater than the third side; difference of any two sides of a triangle is less than the third side.

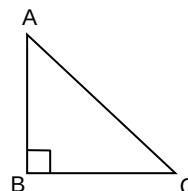
If the sides are arranged in the ascending order of their measurement, the angles opposite the sides (in the same order) will also be in ascending order (i.e., greater angle has greater side opposite to it); if the sides are arranged in descending order of their measurement, the angles opposite the sides in the same order will also be in descending order (i.e., smaller angle has smaller side opposite to it).

There can be only one right angle or only one obtuse angle in any triangle. There can also not be one right angle and an obtuse angle both present at the same time in a triangle.

Hypotenuse is the side opposite the right angle in a right-angled triangle. In a right-angled triangle, the hypotenuse is the longest side. In an obtuse angled triangle, the side opposite the obtuse angle is the longest side.

Fig. 4.08

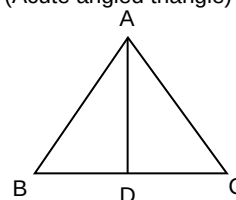
(Right-angled triangle)



In a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. In Fig. 4.08, $AC^2 = AB^2 + BC^2$

Fig. 4.09

(Acute angled triangle)

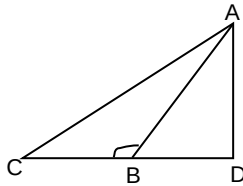


In an acute angled triangle, the square of the side opposite the acute angle is less than the sum of the squares of the other two sides by a quantity equal to twice the product of one of these two sides and the projection of the second side on the first side.

In Fig. 4.09, $AC^2 = AB^2 + BC^2 - 2 \cdot BC \cdot BD$

Fig. 4.10

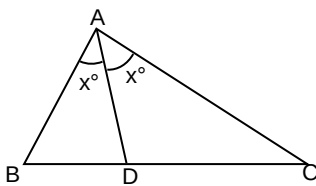
(Obtuse angled triangle)



In an obtuse angled triangle, the square of the side opposite the obtuse angle is greater than the sum of the squares of the other two sides by a quantity equal to twice the product of one of the sides containing the obtuse angle and the projection of the second side on the first side.

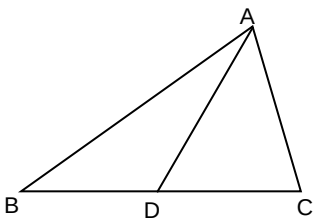
In Fig. 4.10, $AC^2 = AB^2 + BC^2 + 2BC \cdot BD$

Fig. 4.11



In a triangle, the internal bisector of an angle bisects the opposite side in the ratio of the other two sides. In triangle ABC, if AD is the angle bisector of angle A, then $BD/DC = AB/AC$. This is called the **Angle Bisector Theorem** (refer to Fig. 4.11).

Fig. 4.12

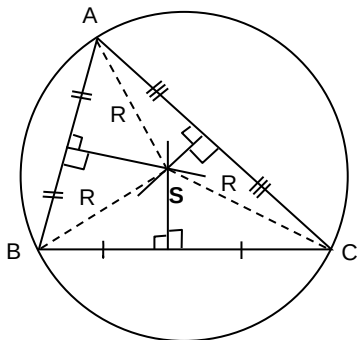


In $\triangle ABC$, if AD is the median from A to side BC (meeting BC at its mid point D), then $2(AD^2 + BD^2) = AB^2 + AC^2$. This is called the **Apollonius Theorem**. This will be helpful in calculating the lengths of the three medians given the lengths of the three sides of the triangle (refer to Fig. 4.12).

GEOMETRIC CENTRES OF A TRIANGLE

CIRCUMCENTRE

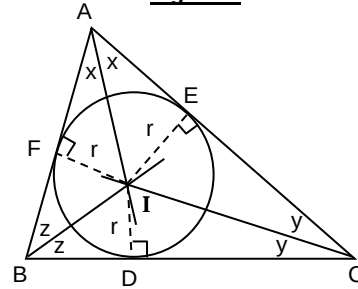
Fig. 4.13



The three perpendicular bisectors of a triangle meet at a point called **Circumcentre** of the triangle and it is represented by S. The circumcentre of a triangle is equidistant from its vertices and the distance of circumcentre from each of the three vertices is called circumradius (represented by R) of the triangle. The circle drawn with the circumcentre as centre and circumradius as radius is called the **Circumcircle** of the triangle and passes through all three vertices of the triangle. (refer to Fig. 4.13)

INCENTRE AND EXCENTRES

Fig. 4.14



The internal bisectors of the three angles of a triangle meet at a point called **Incentre** of the triangle and it is represented by I. Incentre is equidistant from the three sides of the triangle i.e., the perpendiculars drawn from the incentre to the three sides are equal in length and this length is called the inradius (represented by r) of the triangle. The circle drawn with the incentre as centre and the inradius as radius is called the incircle of the triangle and it touches all three sides on the inside.

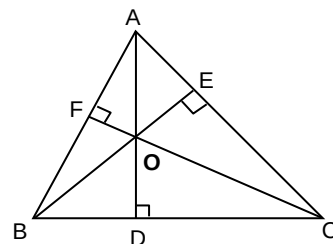
In Fig. 4.14, $\angle BIC = 90^\circ + \frac{1}{2} A$ where I is the incentre.

$\angle CIA = 90^\circ + \frac{1}{2} B$; and $\angle AIB = 90^\circ + \frac{1}{2} C$.

If the internal bisector of one angle and the external bisectors of the other two angles are drawn, they meet at a point called **Excentre**. There will be totally three excentres for the triangle - one corresponding to the internal bisector of each angle.

ORTHOCENTRE

Fig. 4.15



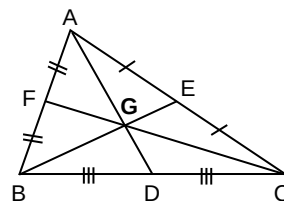
The three altitudes meet at a point called Orthocentre and it is represented by O (refer to Fig. 4.15).

$\angle BOC = 180^\circ - \angle A$; $\angle COA = 180^\circ - \angle B$;

$\angle AOB = 180^\circ - \angle C$.

CENTROID

Fig. 4.16



The three medians of a triangle meet at a point called the **Centroid** and it is represented by G (refer to Fig. 4.16).

Important points about geometric centres of a triangle

In an acute angled triangle, the circumcentre lies inside the triangle. In a right-angled triangle, the circumcentre lies on the hypotenuse of the triangle (it is the midpoint of the hypotenuse). In an obtuse angled triangle, the circumcentre lies outside the triangle.

In an acute angled triangle, the orthocentre lies inside the triangle. In a right-angled triangle, the vertex where the right angle is formed (i.e., the vertex opposite the hypotenuse) is the orthocentre. In an obtuse angled triangle, the orthocentre lies outside the triangle.

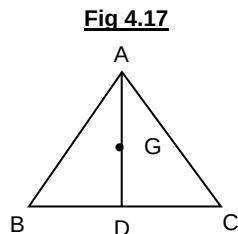
In a right-angled triangle the length of the median drawn to the hypotenuse is equal to half the hypotenuse. This median is also the circumradius of the right-angled triangle.

Centroid divides each of the medians in the ratio 2 : 1, the part of the median towards the vertex being twice in length to the part towards the side.

Inradius is less than half of any of the three altitudes of the triangle.

In an isosceles triangle, the centroid, the orthocentre, the circumcentre and the incentre, all lie on the median to the base.

In an equilateral triangle, the centroid, the orthocentre, the circumcentre and the incentre, all coincide.



Hence, in equilateral triangle ABC shown in Fig. 4.17, AD is the median, altitude, angle bisector and perpendicular bisector. G is the centroid which divides the median in the ratio 2 : 1. Hence, $AG = 2/3 AD$ and $GD = 1/3 AD$.

But since AD is also the perpendicular bisector and angle bisector and since G is the circumcentre as well as the incentre, AG will be the circumradius and GD will be the inradius of the equilateral triangle ABC. Since AD is also the altitude, its length is equal to $\sqrt{3}a/2$ where a is the side of the equilateral triangle. Hence, the circumradius of the equilateral triangle

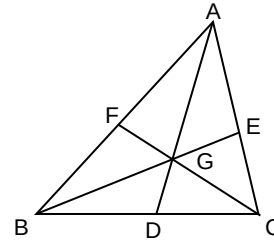
$$= \frac{2}{3} \times \frac{\sqrt{3}}{2} \cdot a = a/\sqrt{3} \text{ and the inradius} = \frac{1}{3} \times \frac{\sqrt{3}}{2} \cdot a$$

$$= a/2\sqrt{3}$$

Since the radii of the circumcircle and the incircle of an equilateral triangle are in the ratio 2 : 1, the areas of the circumcircle and the incircle of an equilateral triangle will be in the ratio 4 : 1.

When the three medians of a triangle (i.e., the medians to the three sides of a triangle from the corresponding opposite vertices) are drawn, the resulting six triangles are equal in area and the area of each of these triangles in turn is equal to one-sixth of the area of the original triangle.

Fig 4.18

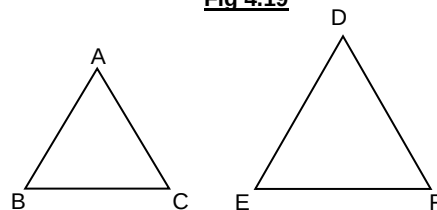


In Fig 4.18, AD, BE and CF are the medians drawn to the three sides. The three medians meet at the centroid G. The six resulting triangles AGF, BGF, BGD, CGD, CGE and AGE are equal in area and each of them is equal to $1/6^{\text{th}}$ of the area of triangle ABC.

SIMILARITY OF TRIANGLES

Two triangles are said to be similar if the three angles of one triangle are equal to the three angles of the second triangle. Similar triangles are alike in shape only. The corresponding angles of two similar triangles are equal but the corresponding sides are only proportional and not equal.

Fig 4.19



For example, in Fig 4.19, if $\triangle ABC$ is similar to $\triangle DEF$ where $\angle A = \angle D$, $\angle B = \angle E$ and $\angle C = \angle F$, we have the ratios of the corresponding sides equal, as given below.

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

By "corresponding sides", we mean that if we take a side opposite to a particular angle in one triangle, we should consider the side opposite to the equal angle in the second triangle. In this case, since AB is the side opposite to $\angle C$ in $\triangle ABC$, and since $\angle C = \angle F$, we have taken DE which is the side opposite to $\angle F$ in $\triangle DEF$.

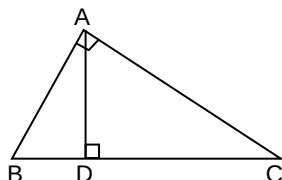
Two triangles are similar if,

- the three angles of one triangle are respectively equal to the three angles of the second triangle, or
- two sides of one triangle are proportional to two sides of the other and the included angles are equal, or
- if the three sides of one triangle are proportional to the three sides of another triangle.

In two similar triangles,

- (a) Ratio of corresponding sides = Ratio of heights (altitudes) = Ratio of the lengths of the medians = Ratio of the lengths of the angular bisectors = Ratio of inradii = Ratio of circumradii = Ratio of perimeters.
- (b) Ratio of areas = Ratio of squares of corresponding sides

Fig 4.20



In a right-angled triangle, the altitude drawn to the hypotenuse divides the given triangle into two similar triangles, each of which is in turn similar to the original triangle. In triangle ABC in Fig.4.20, ABC is a right-angled triangle where $\angle A$ is a right angle. AD is the perpendicular drawn to the hypotenuse BC. The triangles ABD, CAD and CBA are similar because of the equal angles given below.

In triangle ABC, $\angle A = 90^\circ$. If $\angle B = \theta$, then $\angle C = 90^\circ - \theta$.

In triangle ABD, $\angle ADB = 90^\circ$. We already know that $\angle B = \theta$; hence $\angle BAD = 90^\circ - \theta$.

In triangle ADC, $\angle ADC = 90^\circ$. We already know that $\angle C = 90^\circ - \theta$; hence $\angle CAD = \theta$.

We can write down the relationship between the sides in these three triangles. The important relationships that emerge out of this exercise are :

1. $AD^2 = BD \cdot DC$;
2. $AB^2 = BC \cdot BD$;
3. $AC^2 = CB \cdot CD$.

CONGRUENCE OF TRIANGLES

Two triangles will be congruent if at least one of the following conditions is satisfied:

Three sides of one triangle are respectively equal to the three sides of the second triangle (normally referred to as the S-S-S rule, i.e., the side-side-side congruence).

Two sides and the included angle of one triangle are respectively equal to two sides and the included angle of the second triangle (normally referred to as the S-A-S rule, i.e., side-angle-side congruence).

Two angles and one side of a triangle are respectively equal to two angles and the corresponding side of the second triangle (normally referred to as the A-S-A rule, i.e., angle-side-angle congruence).

Two right-angled triangles are congruent if the hypotenuse and one side of one triangle are respectively equal to hypotenuse and one side of the second right-angled triangle.

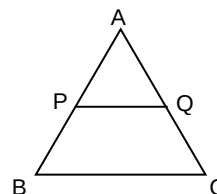
In two congruent triangles,

- the corresponding sides (i.e., sides opposite to corresponding angles) are equal.
- the corresponding angles (angles opposite to corresponding sides) are equal.
- the areas of the two triangles will be equal.

Some more useful points about triangles

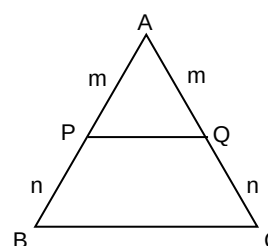
BASIC PROPORTIONALITY THEOREM

Fig. 4.21



A line drawn parallel to one side of a triangle divides the other two sides in the same proportion. For example, in Fig. 4.21, PQ is drawn parallel to BC in $\triangle ABC$. This will divide the other two sides AB and AC in the same ratio, i.e., $AP/PB = AQ/QC$.

Fig. 4.22



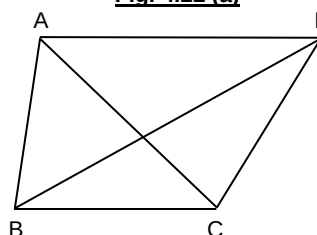
Conversely, a line joining two points (each) dividing two sides of a triangle in the same ratio is parallel to the third side. In Fig. 4.22, P divides AB in the ratio $m : n$ and Q divides AC in the ratio $m : n$. Now, the line joining P and Q will be parallel to the third side BC and the length of PQ will be equal to $\frac{m}{m+n}$ times the length of BC.

We can say that a line drawn through a point on a side of the triangle parallel to a second side will cut the third side in the same ratio as the first side is divided.

MID-POINT THEOREM

The line joining the midpoints of two sides of a triangle is parallel to the third side and it is half the third side.

Fig. 4.22 (a)



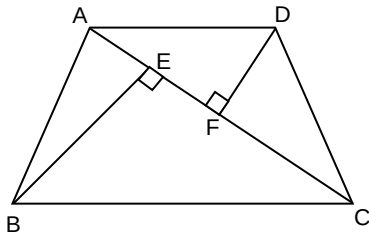
Two triangles having the same base and lying between the same pair of parallel lines have their areas equal (fig.4.22(a)).

AD is parallel to BC. Hence, $\triangle ABC = \triangle DCB$

QUADRILATERALS

Any four-sided closed figure is called a quadrilateral. By imposing certain conditions on the sides and/or angles of a quadrilateral, we can get the figures trapezium, parallelogram, rhombus, rectangle, square.

Fig. 4.23

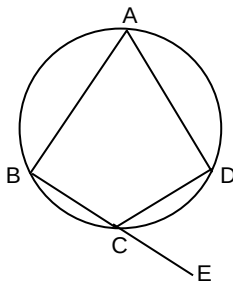


The sum of four angles of a quadrilateral is equal to 360° .

The perpendiculars drawn to a diagonal (in a quadrilateral) from the opposite vertices are called "offsets". In Fig. 4.23, BE and DF are the offsets drawn to the diagonal AC.

If the four vertices of a quadrilateral lie on the circumference of a circle (i.e., if the quadrilateral can be inscribed in a circle) it is called a cyclic quadrilateral (refer to Fig. 4.24). In a cyclic quadrilateral, sum of opposite angles = 180° i.e., in Fig. 4.24, $A + C = 180^\circ$ and $B + D = 180^\circ$.

Fig. 4.24

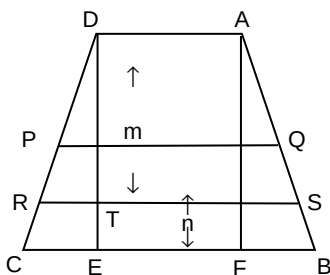


Also, in a cyclic quadrilateral, exterior angle is equal to the interior opposite angle, i.e., in Fig. 4.24, $\angle DCE$ is equal to $\angle BAD$.

Now, we will look at different quadrilaterals and their properties.

TRAPEZIUM

Fig. 4.25



If one side of a quadrilateral is parallel to its opposite side, then it is called a trapezium. The two sides other than the parallel sides in a trapezium are called the oblique sides.

In Fig. 4.25, ABCD is a trapezium where AD is parallel to BC.

If the midpoints of the two oblique sides are joined, it is equal in length to the average of the two parallel sides, i.e., in Fig. 4.25, $PQ = \frac{1}{2} [AD + BC]$

In general, if a line is drawn in between the two parallel sides of the trapezium such that it is parallel to the

parallel sides and also divides the distance between the two parallel sides in the ratio $m : n$ (where the portion closer to the shorter of the two parallel sides is m), the length of the line is given by :

$$\frac{m}{m+n} \times \text{Longer side} + \frac{n}{m+n} \times \text{Shorter side where}$$

shorter side and longer side refer to the shorter and longer of the two parallel sides of the trapezium.

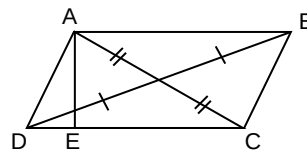
In Fig. 4.25, RS is the line parallel to AD and BC and the ratio of the distances DT and TE is $m : n$.

The length of RS is given by

$$\frac{m}{m+n} \times BC + \frac{n}{m+n} \times AD$$

PARALLELOGRAM

Fig. 4.26



A quadrilateral in which opposite sides are parallel is called a parallelogram.

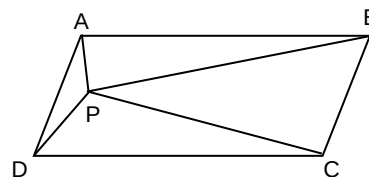
In a parallelogram

- Opposite sides are equal
 - Opposite angles are equal
 - Sum of any two adjacent angles is 180°
 - Each diagonal divides the parallelogram into two congruent triangles.
 - The diagonals bisect each other.
- Conversely, if in a quadrilateral
- the opposite sides are equal or
 - the opposite angles are equal or
 - the diagonals bisect each other or
 - a pair of opposite sides are parallel and equal
- such a quadrilateral is a parallelogram.

If two adjacent angles of a parallelogram are equal, then all four angles will be equal and each in turn will be equal to 90° . Then the figure will be a rectangle.

If any two adjacent sides of a parallelogram are equal, then all four sides will be equal to each other and the figure will be a rhombus.

Fig. 4.27



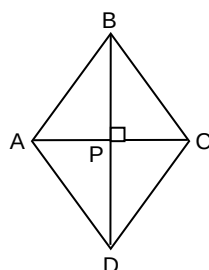
If any point inside a parallelogram is taken and is joined to all the four vertices the four resulting triangles will be such that the sum of the areas of opposite triangles is equal. In Fig. 4.27, P is a point inside the parallelogram ABCD and it is joined to the four vertices of the parallelogram by the lines PA, PB, PC and PD respectively. Then Area of triangle PAB + Area of triangle PCD = Area of triangle PBC + Area of triangle PAD = Half the area of parallelogram ABCD.

If there is a parallelogram and a triangle with the same base and between the same parallel lines, then the area of the triangle will be half that of the parallelogram.

If there is a parallelogram and a rectangle with the same base and between the same parallel lines, then the areas of the parallelogram and the rectangle will be the same. The figure formed by joining the midpoints of the sides of any quadrilateral taken in order, is a parallelogram.

RHOMBUS

Fig. 4.28



A rhombus is a parallelogram in which a pair of adjacent sides are equal (all four sides of a rhombus are equal). Since a rhombus is a parallelogram, all the properties of a parallelogram apply to a rhombus. Further, in a rhombus, the diagonals bisect each other **perpendicularly**.

Conversely, any quadrilateral where the two diagonals bisect each other at right angles will be a rhombus.

The four triangles that are formed by the two bisecting diagonals with the four sides of the rhombus will all be congruent. In Fig. 5.28, the four triangles PAB, PBC, PCD and PAD are congruent.

Side of a rhombus

$$= \sqrt{1/4 \times \text{Sum of squares of the diagonals.}}$$

RECTANGLE

A rectangle also is a special type of parallelogram and hence all properties of a parallelogram apply to rectangles also. A rectangle is a parallelogram in which two adjacent angles are equal or each of the angles is equal to 90° .

The diagonals of a rectangle are **equal** (and, of course, bisect each other).

When a rectangle is inscribed in a circle, the diagonals become the diameters of the circle.

If a and b are the two adjacent sides of a rectangle, then the diagonal is given by $\sqrt{a^2 + b^2}$.

If a rectangle and a triangle are on the same base and between the same parallels, then the area of the triangle will be equal to half the area of the rectangle.

SQUARE

A square is a rectangle in which all four sides are equal (or a rhombus in which all four angles are equal, i.e., all are right angles) Hence, the diagonals are equal and they bisect at right angles. So, all the properties of a rectangle as well as those of a rhombus hold good for a square.

$$\text{Diagonal} = \sqrt{2} \times \text{Side}$$

When a square is inscribed in a circle, the diagonals become the diameters of the circle.

When a circle is inscribed in a square, the side of the square is equal to the diameter of the circle.

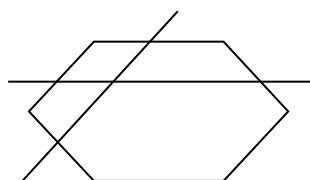
The largest rectangle that can be inscribed in a given circle will be a square.

POLYGON

Any closed figure with three or more sides is called a polygon.

A convex polygon is one in which each of the interior angles is less than 180° . It can be noticed that any straight line drawn cutting a convex polygon passes only two sides of the polygon, as shown in the figure below.

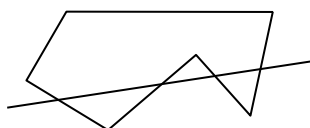
Fig. 4.28 (a)



Convex Polygon

In a concave polygon, it is possible to draw lines passing through more than two sides, as shown in the figure below.

Fig. 4.28 (b)



Concave Polygon

A regular polygon is a convex polygon in which all sides are equal and all angles are equal. A regular polygon can be inscribed in a circle. The centre of the circumscribing circle (the circle in which the polygon is inscribed) of a regular polygon is called the centre of the polygon.

The names of polygons with three, four, five, six, seven, eight, nine and ten sides are respectively triangle, quadrilateral, pentagon, hexagon, heptagon, octagon, nonagon and decagon.

The sum of interior angles of a convex polygon is equal to $(2n - 4)$ right angles where n is the number of the sides of the polygon.

If each of the sides of a convex polygon is extended, the sum of the external angles thus formed is equal to 4 Right Angles (i.e., 360°).

In a regular polygon of n sides, if each of the interior angles is d° , then $d = \frac{2n - 4}{n} \times 90^\circ$ and each exterior

$$\text{angle} = \frac{360^\circ}{n}.$$

It will be helpful to remember the interior angles of the following regular polygons:

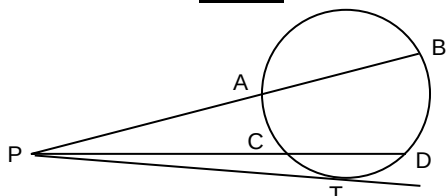
Regular pentagon	: 108°
Regular hexagon	: 120°
Regular octagon	: 135°

If the centre of a regular polygon (with n sides) is joined with each of the vertices, we get n identical triangles inside the polygon. In general, all these triangles are isosceles triangles. Only in case of a regular hexagon, all these triangles are equilateral triangles, i.e., in a regular hexagon, the radius of the circumscribing circle is equal to the side of the hexagon.

A line joining any two non-adjacent vertices of a polygon is called a diagonal. A polygon with n sides will have $\frac{n(n-3)}{2}$ diagonals.

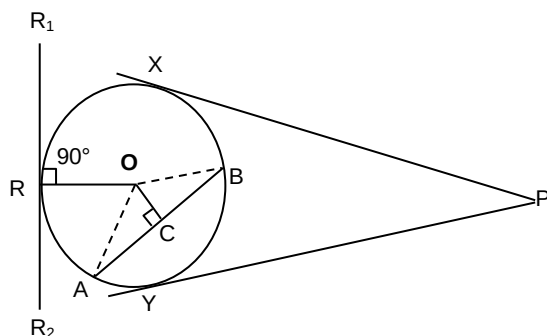
CIRCLES

Fig 4.29



A circle is a closed curve drawn such that any point on the curve is equidistant from a fixed point. The fixed point is called the centre of the circle and the distance from the centre to any point on the circle is called the radius of the circle.

Fig. 4.30



Diameter is a straight line passing through the centre of the circle and joining two points on the circle. A circle is symmetric about any diameter.

A chord is a line joining two points on the circumference of a circle (AB in Fig. 4.30). Diameter is the largest chord in a circle.

A secant is a line intersecting a circle in two distinct points and extending outside the circle also.

A line that touches the circle at only one point is a tangent to the circle (R_1R_2 is a tangent touching the circle at the point R in Fig. 5.30).

If PAB and PCD are two secants (in Fig. 4.29), then $PA \cdot PB = PC \cdot PD$

If PAB and PCD are secants and PT is a tangent to the circle at T (in Fig. 4.29), then $PA \cdot PB = PC \cdot PD = PT^2$.

Two tangents can be drawn to the circle from any point outside the circle and these two tangents are equal in length. In Fig. 4.30, P is the external point and the two tangents PX and PY are equal.

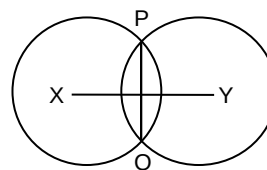
A tangent is perpendicular to the radius drawn at the point of tangency (In Fig. 4.30, $R_1R_2 \perp OR$). Conversely, if a perpendicular is drawn to the tangent at the point of tangency, it passes through the centre of the circle.

A perpendicular drawn from the centre of the circle to a chord bisects the chord (In Fig. 4.30, OC, the perpendicular from O to the chord AB bisects AB) and conversely, the perpendicular bisector of a chord passes through the centre of the circle.

Two chords that are equal in length will be equidistant from the centre, and conversely two chords which are equidistant from the centre of the circle will be of the same length.

One and only one circle passes through any three given non-collinear points.

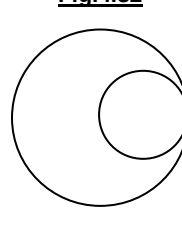
Fig. 4.31



When there are two intersecting circles, the line joining the centres of the two circles will perpendicularly bisect the line joining the points of intersection. In Fig. 4.31, the two circles with centres X and Y respectively intersect at the two points P and Q. The line XY (the line joining the centres) bisects PQ (the line joining the two points of intersection).

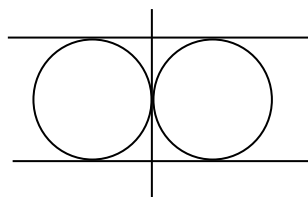
Two circles are said to touch each other if a common tangent can be drawn touching both the circles at the same point. This is called the point of contact of the two circles. The two circles may touch each other internally (as in Fig. 4.32) or externally (As in Fig. 4.33). When two circles touch each other, then the point of contact and the centres of the two circles are collinear, i.e., the point of contact lies on the line joining the centres of the two circles.

Fig.4.32



If two circles touch internally, the distance between the centres of the two circles is equal to the difference in the radii of the two circles. When two circles touch each other externally, then the distance between the centres of the two circles is equal to the sum of the radii of the two circles.

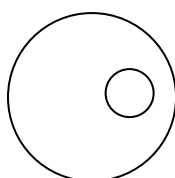
Fig. 4.33



A tangent drawn common to two circles is called a common tangent. In general, for two circles, there can be anywhere from zero to four common tangents drawn depending on the position of one circle in relation to the other.

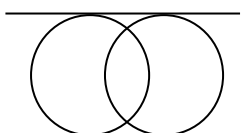
If the common tangent is either parallel to the line of centres or cuts the line joining the centres not between the two circles but on one side of the circles, such a common tangent is called a direct common tangent. A common tangent that cuts the line joining the centres in between the two circles is called transverse common tangent.

Fig. 4.34



If two circles are such that one lies completely inside the other (without touching each other), then there will not be any common tangent to these circles (refer to Fig. 4.34). Two circles touching each other internally (i.e., still one circle lies inside the other), then there is only one common tangent possible and it is drawn at the point of contact of the two circles (refer to Fig. 4.32).

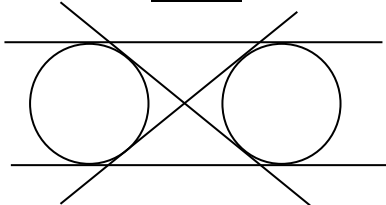
Fig. 4.35



Two intersecting circles have two common tangents. Both these are direct common tangents and the two intersecting circles do not have a transverse common tangent (refer to Fig. 4.35).

Two circles touching each other externally have three common tangents. Out of these, two are direct common tangents and one is a transverse common tangent. The transverse common tangent is at the point of contact (Refer to Fig. 4.33).

Fig. 4.36



Two circles which are non-intersecting and non-enclosing (i.e. one does not lie inside the other) have four common tangents - two direct and two transverse common tangents (Refer to Fig. 4.36).

If r_1 and r_2 are the radii of the two non-intersecting non-enclosing circles,

Length of the direct common tangent

$$= \sqrt{(\text{Distance between centre})^2 - (r_1 - r_2)^2}$$

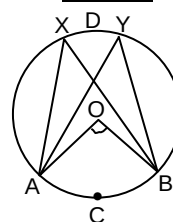
Length of transverse common tangent

$$= \sqrt{(\text{Distance between centre})^2 - (r_1 + r_2)^2}$$

Two circles are said to be concentric if they have the same centre. As is obvious, here the circle with smaller radius lies completely within the circle with bigger radius.

Arcs and Sectors

Fig. 4.37



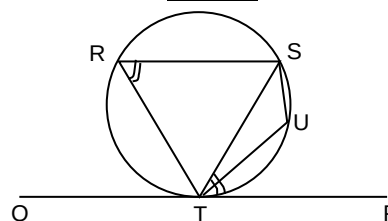
An arc is a segment of a circle. In Fig. 4.37, ACB is called minor arc and ADB is called major arc. In general, if we talk of an arc AB, we refer to the minor arc. AOB is called the angle formed by the arc AB (at the centre of the circle).

The angle subtended by an arc at the centre is double the angle subtended by the arc in the remaining part of the circle.

In Fig. 4.37, $\angle AOB = 2 \cdot \angle AXB = 2 \cdot \angle AYB$

Angles in the same segment are equal. In Fig. 4.37, $\angle AXB = \angle AYB$.

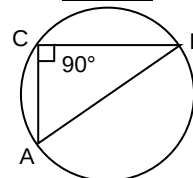
Fig. 4.38



The angle between a tangent and a chord through the point of contact of the tangent is equal to the angle made by the chord in the alternate segment (i.e., segment of the circle on the side other than the side of location of the angle between the tangent and the chord). This is normally referred to as the "alternate segment theorem." In Fig. 4.38, PQ is a tangent to the circle at the point T and TS is a chord drawn at the point of contact. Considering $\angle PTS$ which is the angle between the tangent and the chord, the angle TRS is the angle in the "alternate segment". So, $\angle PTS = \angle TRS$.

Similarly, $\angle QTS = \angle TUS$.

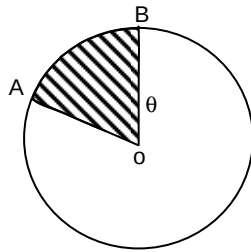
Fig. 4.39



We have already seen in quadrilaterals, the opposite angles of a cyclic quadrilateral are supplementary and that the external angle of a cyclic quadrilateral is equal to the interior opposite angle.

The angle in a semicircle (or the angle the diameter subtends in a semicircle) is a right angle. The converse of the above is also true and is very useful in a number of cases - in a right angled triangle, a semi-circle with the hypotenuse as the diameter can be drawn passing through the third vertex (Refer to Fig. 4.39).

Fig. 4.40



The area formed by an arc and the two radii at the two end points of the arc is called sector.

In Fig. 4.40, the shaded figure AOB is called the minor sector.

AREAS OF PLANE FIGURES

Mensuration is the branch of geometry that deals with the measurement of length, area and volume. We have looked at properties of plane figures till now. Here, in addition to areas of plane figures, we will also look at surface areas and volumes of "solids." Solids are objects, which have three dimensions (plane figures have only two dimensions).

Let us briefly look at the formulae for areas of various plane figures and surface areas and volumes of various solids.

TRIANGLES

The area of a triangle is represented by the symbol Δ . For any triangle, the three sides are represented by a, b and c and the angles opposite these sides represented by A, B and C respectively.

- (i) For any triangle in general,
- (a) When the measurements of three sides a, b, c are given,

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$
where $s = \frac{a+b+c}{2}$
This is called Hero's formula.
- (b) When base (b) and altitude (height) to that base are given,

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{altitude} = \frac{1}{2} b.h$$
- (c)
$$\text{Area} = \frac{1}{2} ab \cdot \sin C = \frac{1}{2} bc \cdot \sin A = \frac{1}{2} ca \cdot \sin B$$
- (d)
$$\text{Area} = \frac{abc}{4R}$$
 where R is the circumradius of the triangle.
- (e)
$$\text{Area} = r.s$$
 where r is the inradius of the triangle and s, the semi-perimeter.

Out of these five formulae, the first and the second are the most commonly used and are also more important from the examination point of view.

- (ii) For a right angled triangle,

$$\text{Area} = \frac{1}{2} \times \text{Product of the sides containing the right angle}$$

- (iii) For an equilateral triangle

$$\text{Area} = \frac{\sqrt{3} \cdot a^2}{4} \text{ where "a" is the side of the triangle}$$

$$\text{The height of an equilateral triangle} = \frac{\sqrt{3} \cdot a}{2}$$

- (iv) For an isosceles triangle

$$\text{Area} = \frac{b}{4} \sqrt{4a^2 - b^2} \text{ where "a" is length of each of the two equal sides and b is the third side}$$

QUADRILATERALS

- (i) For any quadrilateral

$$\text{Area of the quadrilateral} = \frac{1}{2} \times \text{One diagonal} \times \text{Sum of the offsets drawn to that diagonal}$$

Hence, for the quadrilateral ABCD shown in Fig. 4.23, area of quadrilateral ABCD = $\frac{1}{2} \times AC \times (BE + DF)$

- (ii) For a cyclic quadrilateral where the four sides measure a, b, c and d respectively,

$$\text{Area} = \sqrt{(s-a)(s-b)(s-c)(s-d)} \text{ where s is the semi-perimeter, i.e., } s = (a+b+c+d)/2$$

- (iii) For a trapezium

$$\begin{aligned} \text{Area of a trapezium} &= \frac{1}{2} \times \text{Sum of parallel sides} \times \text{Distance between them} \\ &= \frac{1}{2} \times (AD + BC) \times AE \text{ (refer to Fig. 4.25)} \end{aligned}$$

- (iv) For a parallelogram

$$(a) \text{ Area} = \text{Base} \times \text{Height}$$

$$(b) \text{ Area} = \text{Product of two sides} \times \text{Sine of included angle}$$

- (v) For a rhombus

$$\text{Area} = \frac{1}{2} \times \text{Product of the diagonals}$$

$$\text{Perimeter} = 4 \times \text{Side of the rhombus}$$

- (vi) For a rectangle

$$\text{Area} = \text{Length} \times \text{Breadth}$$

$$\text{Perimeter} = 2(l + b), \text{ where l and b are the length and the breadth of the rectangle respectively}$$

- (vii) For a square

$$(a) \text{ Area} = \text{Side}^2$$

$$(b) \text{ Area} = \frac{1}{2} \times \text{Diagonal}^2$$

$$[\text{where the diagonal} = \sqrt{2} \times \text{side}]$$

$$\text{Perimeter} = 4 \times \text{Side}$$

- (viii) For a regular hexagon

$$(a) \text{ Area} = \frac{3\sqrt{3}}{2} (\text{side})^2$$

$$(b) \text{ Perimeter} = 6(\text{side})$$

- (ix) For a polygon

$$(a) \text{ Area of a regular polygon} = \frac{1}{2} \times \text{Perimeter} \times \text{Perpendicular distance from the centre of the polygon to any side}$$

(Please note that the centre of a regular polygon is equidistant from all its sides)

- (b) For a polygon which is not regular, the area has to be found out by dividing the polygon into suitable number of quadrilaterals and triangles and adding up the areas of all such figures present in the polygon.

CIRCLE

- (i) **Area of the circle** = πr^2 where r is the radius of the circle

$$\text{Circumference} = 2\pi r$$

- (ii) Sector of a circle

$$\text{Length of arc} = \frac{\theta}{360^\circ} \times 2\pi r$$

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 \text{ where } \theta \text{ is the angle of the}$$

sector in degrees and r is the radius of the circle.

Area = $(1/2)lr$; l is length of arc and r is radius.

- (iii) Ring : Ring is the space enclosed by two concentric circles.

Area = $\pi R^2 - \pi r^2 = \pi(R + r)(R - r)$ where R is the radius of the outer circle and r is the radius of the inner circle.

ELLIPSE

Area = πab where " a " is semi-major axis and " b " is semi-minor axis.

Perimeter = $\pi(a + b)$

